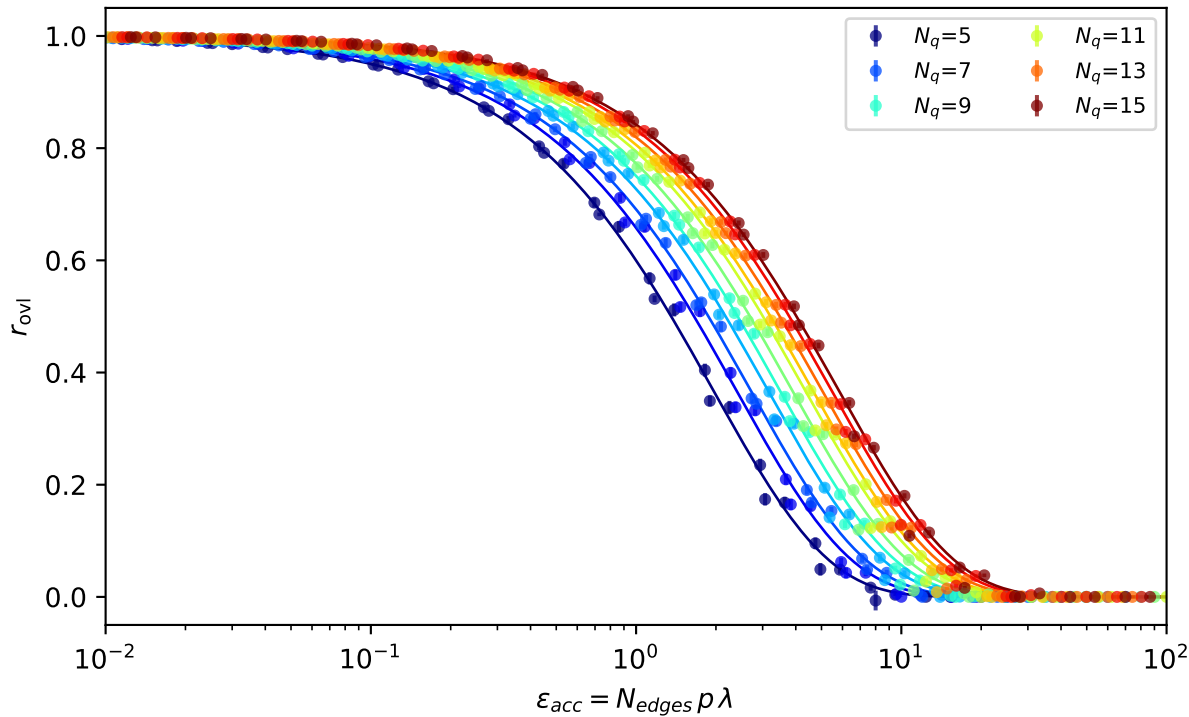


approximation-ratio overlap, all depths

 $\kappa_r$  against chain length